BWB Technologies

Just Add Gas

inkarp®
Your Knowledge... Our Solution...!

BWB XP Flame Photometer



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Flame photometers that push the boundaries of low temperature flame analysis.

Analysis with a low temperature propane, LPG, or butane flame has been redefined by BWB Technologies.







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Our Mission

To redefine the boundaries of Flame Emission Spectroscopy (FES) by supplying unique, cost effective, reliable and accurate flame photometers (and related products) and to build strong, lasting relationships with our customers by providing the best customer service in the industry.

About BWB Technologies

BWB Technologies is a UK company focused on the design, manufacture and sales of award winning flame photometers. Drawing on an Anglo-American team of leading industry specialists, we strive to create high quality, cost effective products that redefine what is achievable with a low temperature flame. From our manufacturing plant in Newbury, England, BWB Technologies have introduced a series of flame photometer products and accessories which exceed existing expectations in terms of specification, usability, accuracy, reporting, build quality and value for money.

International support network

BWB Technologies are the only flame photometer manufacturer to have dedicated sales and technical support offices in Europe, The America's and the Middle East. We offer an unsurpassed level of customer support with a 24/7 support service and real people answering customer calls.

Pushing the boundaries of flame photometry

What can be achieved with a flame photometer has been redefined by BWB Technologies. With unrivalled accuracy, low cost and ease of use the BWB flame photometer is not only the best flame photometer available but is the first real alternative to AAS (atomic absorption spectrometry) and ICP (Inductively Coupled Plasma) for measuring Lithium (Li), Sodium (Na), Potassium (K), Calcium (Ca), Caesium (Cs) and Barium (Ba). The recent introduction of industry specific products for Biological samples, Sugar, Nuclear and others further defines BWB Technologies as the leader in "next generation" flame photometers.

Best customer service in the industry

- International telephone service available 24/7, with local access numbers in Europe, the USA and the Middle East
- An international distributor network servicing the needs of customers in over 100 countries.
- Fast delivery of spare parts and consumables backed by extensive inventory in multiple strategic locations.
- Pre-sales application specific product selection assistance
- 100's of hours of video training available in a wide choice of languages
- Technically competent customer service staff
- ISO9001 Certified manufacturer
- Excellent Globally acclaimed after-sale support and service by Value Added Distributors and Internet based assistance from GMT and PST time zones.

Contact Information

BWB Technologies UK Ltd Unit 7, The Galloway Centre, Hambridge Lane, Newbury, RG14 5TL UK

T Europe + 44 (0)1787 273 451 United States 1 (800) 608-9870 Middle East +357 22 666080

E Orders orders@bwbtech.com Admin admin@bwbtech.com Technical technicalsupport@bwbtech.com Sales sales@bwbtech.com

www.bwbtech.com 12 months warranty

BWB Technologies UK, Ltd. warrants, subject to the conditions itemised within this document, through either BWB personnel or personnel of its authorised distributors, to repair or replace free of all charges, including labour, any part of this product which fails within the warranty time specified above, appertaining to this particular product. Such failure must have occurred because of a defect in material or workmanship and not have occurred as a result of operation of the product other than in accordance with procedures described in the instructions furnished with this product.

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Common Features

All BWB products are built around the same innovative product design and world leading technologies, sharing the same core features across the range.

Just add gas!
BWB have revolutionised the flame photometry market by offering an instrument with all accessories supplied as standard. This means that the process of ordering a BWB flame photometer is incredibly simple. There are no complex product codes, accessory lists or ordering requirements. When you order a BWB flame photometer it will arrive with all the accessories you need.

Up to 5 Channels of simultaneous detection and display
The ability to measure and calibrate up to 5 different ions at the same time offer huge advantages over more traditional instrumentation. As the BWB is the first and only flame photometer to offer simultaneous detection and display, our customers do not need to waste valuable time changing filters and re-calibrating when a different measurement is required.

Built-in air compressor
Greatly reduces noise and laboratory footprint while increasing instrument stability, accuracy and reproducibility.

Advanced calibration system
The ability to store calibration data and perform multi-mode calibrations reduces analysis time significantly and increases accuracy.

Interfaces
Connect to your PC using USB 2. A 0-2.5volt output is supplied as standard for a chart recorder or for process control a 4-20mA output is available.

New modern technology offers increased accuracy and reliability
Introduced in 2005, the BWB flame photometer was designed from the ground up, using new, innovative and modern technology to give unsurpassed levels of accuracy, usability and reliability. With clean lines and contemporary bright colour schemes, the instrument looks exceptionally modern and will fit into any laboratory. The instrument has a large LCD display which can be easily viewed from any angle.

Computer connectivity as standard!
The FP-PC software package, included with every purchase, has been written with GLP compliance in mind and facilitates the creation of data and reports in .pdf and .csv formats.

Ease of use
With exceptional ease of use, a clear backlit liquid crystal display guides users through analysis using simple menu options. The display, used in conjunction with a comprehensive membrane switch panel ensures that accurate and reliable results can be obtained with minimal training.

As modern businesses may experience a large turnover of staff, laboratory managers will appreciate the comprehensive training video supplied with every BWB flame photometer. Our training videos are also available online in over 10 languages at www.bwbtech.com and on BWB's YouTube Channel http://www.youtube.com/user/BWBTechnologiesUKLtd

Long term savings
Excellent value for money and very low total cost of ownership.

Selective analysis Flame photometry excludes most elements which means very little interference.

Reliable & simple
Manipulation of the flame is the only adjustment available to the operator during operation ensuring reliable and accurate measurements every time.

Fast!
Once the instrument has stabilized it takes just a matter of seconds to obtain results.

Collection cup
Optional for 'online', continuous measurements, the collection cup turns the BWB Flame Emission Spectrometer (FES) into an 'online', real time, 24/7 analyser with 4-20mA output linked to a single channel. Via a 2 wire output the BWB FP can seamlessly integrate with any SCADA system. Totally unique and available on all models.



BWB Technologies

Li 3 Na 11 K 19 Ca 20 Ba 56 XP

The Award Winning **BWB XP**



A five channel digital Flame Photometer

Ready to use straight out of the box! General Purpose Flame Photometer

The BWB XP is the first and only 5 channel flame photometer with simultaneous detection and display of all 5 elements.

The BWB XP was designed from the ground up, using new, innovative and modern technology to give unsurpassed levels of accuracy, usability and reliability while significantly reducing analysis time. Our customers do not need to waste valuable time changing filters and re-calibrating when a different measurement is required, as is a common requirement in all other Flame Photometer designs from 1951 to the present.

With clean lines and contemporary bright colour schemes, the XP looks exceptionally modern and will fit into any laboratory. The instrument has a large LCD display, which can be viewed easily from any angle.

Our metal casework provides a sturdy portable instrument and enables future recycling opportunities. ROHS compliant components and materials are used wherever possible. The BWB Carbon Footprint is exceptionally low, in keeping with corporate objectives of being a responsible manufacturer.

Unlike other manufacturers who charge extra for vital accessories, BWB include everything required to use the product straight out of the box. Just add Gas!! (Propane, LPG or butane)

We also provide as standard, data sharing via USB from the BWB XP to your PC. This ties the output of the BWB Flame Emission Spectrometer (FES) to the Internet or your company intranet systems and facilitates Distributed Control which is vital for Multi-National companies, Universities and various Governmental agencies.

The BWB XP is the greatest advance in flame photometry for 30 years!







+ FEATURES

- SIMULTANEOUS DETECTION AND DISPLAY OF ALL 5 ATOMS OF INTEREST
- INTUITIVE USER INTERFACE FOR TRUE EASE OF USE
- DISPLAY PROMPTS STEP BY STEP OPERATION
- BUILT-IN AIR COMPRESSOR
- SOLUTIONS AND LABWARE INCLUDED
- DATA SHARING VIA PC LINK VIA USB
- OPERATOR INDEPENDENT DETERMINATION OF RESULTS
- 5 USER SELECTABLE UNITS OF MEASURE
- USER SELECTABLE DECIMAL PLACES
- INTEGRATED PRINTER OPTION USES READILY AVAILABLE PAPER
- IQ, OQ, PQ WEB-BASED CERTIFICATION AVAILABLE: IQ OQ PQ is completely unique in the analytical instrumentation arena. Available in English, Russian, German, Mandarin, French and Spanish. More languages in development.
- CORRECTION OF CA FOR THE INTERFERENCE FROM HIGH LEVELS OF NA

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AFHS

BWB AFHS and Printer



Simple, robust & reliable Automated Fluid Handling System that works seamlessly with BWB's range of Flame Photometers.

BWB customers rely on our instrumentation that may run throughout the day processing tens or even hundreds of samples. Until now this operation has been a manual process and requires an operator present to introduce each sample. Many organisations are striving for greater efficiency and automation of their laboratory processes and in response BWB have a solution. The new Automatic Fluid Handling System (AFHS), which for the first time offers fast, reliable and effective automation of the flame photometry process.

New "Auto Read" Technology is the foundation for automation

The goal of an automated system is to perform tasks accurately, reliably and autonomously without the need for human intervention. By clever software/firmware handling of the input from our multiple channel detectors, stability of readings are analytically determined and no longer a subjective user decision.

Printer

This ultra reliable thermal printer sits atop the main flame photometer enclosure for footprint reduction in valuable lab space. The printer is supplied with print stock derived from recycled paper, has a 10 year shelf life and comes with soy based ink as standard. The dimensions of the unit were also designed to offer compatibility with standard calculator print paper available globally from any high street.

Autosampler

The BWB auto sampler is available as an optional extra for all the latest models of the BWB flame photometer.

Autodiluter

BWB has introduced an operator configurable 1:6 to 1:100 auto-diluter for the latest models of the BWB flame photometer.







+ FEATURES

- BUILT AROUND BWB'S PROPRIETARY "AUTO READ" TECHNOLOGY
- "IRS" (INTERNAL REFERENCE STANDARD) AVAILABLE
- COMPATIBLE WITH ALL 2014 AND ONWARDS BWB FLAME PHOTOMETER MODELS
- 5 USER SELECTABLE TRAY CALIBRATION CORRECTION POINTS
- USER SELECTABLE DILUTION RATIO
- PURPOSE BUILT AUTO DILUTER
- ON DEMAND PRINTING OF KEY ANALYSIS DATA WITH BWB'S INTEGRATED PRINTER
- A SINGLE SHARED POWER SOURCE FOR THE AUTO SAMPLER AND AUTO DILUTER
- ADDITIONAL SAMPLE TRAYS AVAILABLE FOR 1ML & 3ML SAMPLE CUPS
- UP TO 89 SAMPLES
- INTERCHANGEABLE SAMPLE TRAY
- LARGE STOCKS OF SAMPLE CUPS HELD FOR SAME DAY DISPATCH TO PREVENT DELAYS

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AFHS

BWB BIO with AFHS Technical Data

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Sample rate
4ml/min

Tubing materials
Translucent PTFE and Tygon®

Amount of fluid used per sample based on dilution ratios,
DR1 - 5ml
DR10 - 0.5ml
DR100 - 0.05ml

Specificity
Na/K/Li = <1% to each other when equal in concentration at <100ppm

Required desk space for instrument
50cm (H) x 65cm (W) x 63cm (D)
*BIO Flame photometer with BWB Recommended set up positioning. For Safety reasons the Flame photometer requires 1m of unobstructed space above to allow dissipation of heat from the chimney.

Instrument size
BIO Flame Photometer
51cm (H) x 39cm (W) x 35cm (D)
(20in x 15in x 14in)
Auto Diluter
30cm (H) x 30cm (W) x 43cm (D)*
(12in x 12in x 17in)
Auto Sampler
30cm (H) x 18cm (W) x 11cm (D)**
(12in x 7in x 4in)
Shipping
65cm (H) x 49cm (W) x 43cm (D)
(25in x 19in x 17in)
*Inclusive of Cannula operating arm.
** Excluding Dilution Reservoir.

Weight
BIO Flame Photometer - 14g
Auto Diluter - 3.75kg
Auto Sampler - 6.9kg
Shipping
BIO Flame Photometer - 25kg
AFHS unit - 17kg

Optimal range
Multi-point calibration
Na - 0.2 - 300mmol/l*
K - 0.2 - 200mmol/*
Li - 0.2 - 20mmol/l*
Ca - 1 - 20mmol/l*
*With required Dilution Ratios

Reproducibility
<1.5% Coefficient of variability for 10 consecutive samples within optimal range. (After instrument stabilisation).

Limit of detection (LOD) and limit of quantification (LOQ) displayed

LOD	LOQ
Na - 0.0013mmol/l	Na - 0.0043mmol/l
K - 0.0005mmol/l	K - 0.0013mmol/l
Li - 0.003mmol/l	Li - 0.0072mmol/l
Ca - 0.0075mmol/l	Ca - 0.025mmol/l

Time to stability
Less than 15 seconds after sample is introduced into the flame

Single sample time
With Auto Diluter: 130 Seconds including Sample Preparation, Acquisition and flush
Without Auto Diluter: 80 Seconds including Acquisition and flush

Amount of samples measured per hour
With Auto Diluter: 25
Without Auto Diluter: 40

Ions measured
Serum Na and K or Li and Ca
Urine Na and K or Li and Ca

Dilution reservoir size
2.5L with inbuilt low level control
Optional 5L tank available

Amount of samples per dilution reservoir
2.5L - 120* 5L - 240*
*dependant on idle times between trays and sample set-ups.

Amount of samples per tray
89 with additional interchangeable trays to speed up sampling time.

Calibration concentrations

URINE

Point 1	Point 2	Point 3
Na 25mmol/l	Na 50mmol/l	Na 100mmol/l
K 25mmol/l	K 50mmol/l	K 100mmol/l
Li 2.5mmol/l	Li 5mmol/l	Li 10mmol/l
Ca 2.5mmol/l	Ca 5mmol/l	Ca 10mmol/l

Point 4	Point 5
Na 200mmol/l	Na 300mmol/l
K 150mmol/l	K 200mmol/l
Li 15mmol/l	Li 20mmol/l
Ca 15mmol/l	Ca 20mmol/l

SERUM

Point 1	Point 2	Point 3
Na 120mmol/l	Na 130mmol/l	Na 140mmol/l
K 3mmol/l	K 4mmol/l	K 5mmol/l
Li 0.5mmol/l	Li 0.5mmol/l	Li 1mmol/l
Ca 1.5mmol/l	Ca 2mmol/l	Ca 2.5mmol/l

Point 4	Point 5
Na 150mmol/l	Na 160mmol/l
K 6mmol/l	K 7mmol/l
Li 1.5mmol/l	Li 2mmol/l
Ca 3mmol/l	Ca 3.5mmol/l

Recommended minimum warm up time based on ambient temperature
21ºC - 25 minutes

Power requirements
100V - 250V AC at 50 or 60Hz

Fuel requirement
Propane, Butane or Natural Gas* regulated to 19Bar. Flow rate of 0.4l/min
*with modifications. BWB Technologies recommends either Propane or Butane for optimum results.

Readout
LCD, four line, alpha numeric, back lit.

+ WHAT'S IN THE BOX

- PC leads USB and RS232
- Power lead to suit region
- Gas hose
- Aspiration kit
- Manuals covering all aspects of the BIO FP and AFHS
- Quick start guides
- Warranty registration form
- Certificates of analysis for all provided fluids
- Material safety data sheets for all provided fluids
- IQ, OQ, PQ on-line documentation. Certificate of compliance will be issued upon completion of the PQ testing.
- The FP-PC software installation CD
- 150ml of BWB recommended cleaning solution
- Complete set of calibration fluids including diluent concentrate
- 1000 x 5ml AFHS sample vials
- 100ml conical flask with screw lid
- 100 disposable sample cups
- 1L of de-Ionised water
- 2m of waste tube
- Set of spare fuses
- 10 x 10ml pipettes
- 10 x 1ml pipettes



Pack contents

BWB Technologies

Li 3 Na 11 K 19 Ca 20

The **BWB Flash**



The newest feature to Flame Photometry

Finally, a Flame Photometer specific to your needs. The first simultaneous Flame Photometer with built in compressor and configurable element analysis to your requirements.

The BWB Flash is the first and only fully configurable up to four channel flame photometer that detects and displays all calibrated ions at the same time.

The BWB Flash was created by combining the award-winning features of our XP model with new, innovative, and cutting-edge technology to provide unrivalled levels of accuracy, usability, and reliability while significantly reducing analysis time. When a different measurement is required, our customers do not need to waste valuable time changing filters and re-calibrating, as is a common requirement in all other Flame Photometer designs from 1951 to the present.

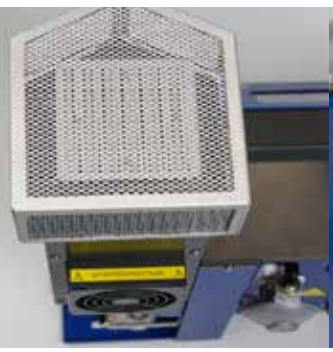
The Flash looks exceptionally modern, with clean lines and contemporary bright colour schemes that will fit into any laboratory. The instrument has a large 5" touch screen display that can be easily viewed from any angle and operated with most laboratory disposable gloves.

Our metal casework creates a sturdy portable instrument with nearly limitless future recycling possibilities. Of course, ROHS-compliant components and materials are used throughout.

The BWB Carbon Footprint is exceptionally low, in keeping with the company's goals of being a responsible manufacturer, and we've succeeded in lowering the gas consumption rate by 13% while also lowering the electricity consumption (kWh) by 26% compared to previous model generations.

Unlike other manufacturers who charge extra for essential accessories, BWB includes everything needed to use the product right away. Simply add Gas!! (Propane, LPG, or butane) We also include data sharing via USB from the BWB Flash to your PC as standard. This connects the BWB Flame Emission Spectrometer (FES) output to the Internet or your company intranet systems, allowing for Distributed Control, which is critical for multinational corporations, universities, and government agencies.

BWB continues to improve our Flame Photometric systems in order to provide the market with unrivalled quality!







+ FEATURES

- LARGE 5" TOUCH SCREEN DISPLAY COMPATIBLE WITH THE MAJORITY OF DISPOSABLE LABORATORY GLOVES
- TRUE EASE OF USE WITH LARGE BUTTONS AND SIMPLE TO UNDERSTAND TEXT AND SYMBOLS
- BUILT-IN AIR COMPRESSOR AND GAS REGULATOR
- CALIBRATION STANDARDS SUPPLIED AS STANDARD IN 10,000PPM CONCENTRATIONS
- AUTO-READ TECHNOLOGY ENABLES OPERATOR INDEPENDENT DETERMINATION OF RESULTS
- USER SELECTABLE DECIMAL PLACES
- IQ, OQ, PQ WEB-BASED CERTIFICATION AVAILABLE: IQ OQ PQ is completely unique in the analytical instrumentation arena. Available in English, Russian, German, Mandarin, French and Spanish. More languages in development.
- COMPLETE SET OF BASIC LABWARE INCLUDED TO GET YOU STARTED

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Flash Models

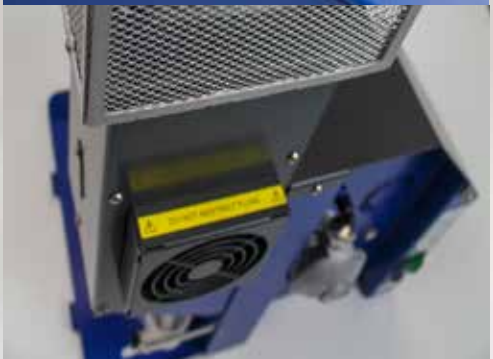
The BWB Flash is the first touchscreen Flame Photometer available in 7 models to suit a wide range of applications and budgets, only pay for the elements you're interested in.

Common features include:

- 10 Point calibration curve
- Internal results storage
- Internal compressor and regulator
- .PDF and .CSV report generation
- IQ OQ PQ possible









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Flash Models



Includes analysis of Sodium and Potassium



Includes analysis of Sodium, Potassium and Lithium



Includes analysis of Sodium, Potassium, Lithium and Calcium



Includes analysis of Calcium



Includes analysis of Potassium



Includes analysis of Lithium



Includes analysis of Sodium

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FP-PC

FP-PC and BWB Drive

FP-PC app

The BWB flame photometer has been configured to easily satisfy the most sophisticated IT requirements. The FP-PC app provides a useful interface to the BWB Flame Photometer (FP) facilitating the following operations:

- Normal calibration Report production
- Logging of streaming data for continuous on-line applications. (A 4-20 mA signal can also be output proportional to a calibrated channel on the FP).
- Extraction of data from the flame photometer when operated in automatic or semi-automatic mode

Calibration manipulation including:
1. Upload from / download to the FP
2. Storage / retrieval of calibrations using the PC or system hard drive

The FP-PC app, included with every purchase, has been written with GLP compliance in mind and facilitates the creation of data and reports in PDF format.

BWB drive

Continuing to building on modern technologies, BWB have created an industry leading USB flash drive packed full of features and resources designed to aide your daily analysis and BWB experience. Exclusively for BWB customers, this self-contained resource includes:

- Tutorial videos in 10 languages
- PC and Mac compatibility
- FP-PC app installation package (PC only)
- Links to BWB's growing online video library (internet connection required)
- Manuals, guides, warranty information and registration PDF
- A list of international support and distributor contact details
- 13 GB of free space for you to use as you please!



www.bwbcustomerportal.com

The BWB Customer Portal is open to join free of charge, BWB Flame Photometer's documentation is all available in one location and is now simpler to access than ever. Keep data under control, access, read, and download the newest documents safely. By staying connected, you can never miss a crucial system update or updates to configuration, standard operating procedures, or simple yet helpful hints and advice, here is a little snippet of what can be found on the Customer Portal:

- Certificate of analysis
- Information and help sheets
- MSDS
- Packing Lists
- Software Installation
- User manuals
- Service manuals
- Certification

IQ OQ PQ Certification

IQ OQ PQ is a unique validation and certification program from BWB.

The programme has been specially created to ensure the end user has the technical knowledge required to install and initialise their BWB Flame Photometer and the technical know-how and confidence to operate any of our 2015 and onwards range of flame photometers correctly and efficiently on a daily basis.

IQ - Installation Qualification

The installation qualification covers in detail the key steps required to correctly set up the BWB range of photometers. IQ stages include: maintenance and service plans, the installation environment and health and safety requirements.

OQ - Operation Qualification

The operation qualification covers the steps required to ensure that the BWB flame photometer is operating correctly.

The user will undertake a series of hardware checks and test readings, providing the results to BWB for analysis.

PQ - Performance Qualification

The performance qualification has been designed to ensure that the instruments perform to the original factory specifications.

Once the IQ and OQ sections have been completed, the user runs a sample manually with the solutions, prints out the results and sends to BWB for analysis.

After completion of the qualification and confirmation from BWB, a signed certificate will be issued to the user.

IQ OQ PQ is a unique and effective program built to ensure technical ability, confidence and peace of mind for all users of the BWB 2015 and onwards Series of Flame Photometer.

To find more about IQ OQ PQ get in touch at info@bwbtech.com

GLP - Compliance

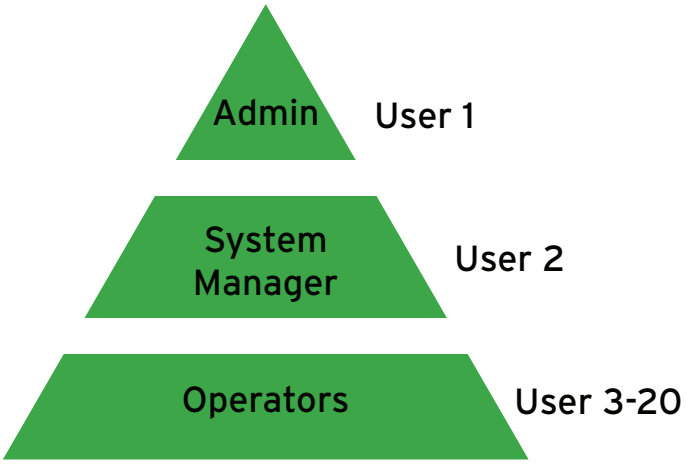
With an increase in demand for access compliant systems we've responded with the development of our Access Control Technology. Available and configured as standard on all new BIO and BIO AV models at no additional cost.

All calibrations and results are tagged with user IDs to indicate who did the calibration / correction / test; the user ID appears on all printouts. Calibrations and results are non-changeable. The facility to download a calibration from the FP-PC Software has been disabled for instruments with User Access Control. The instrument may still be connected to a PC running BWB FP-PC Software with all the other functions still operational.

User 1 is System Admin and has full access to entire system including setting all user pass codes.

User 2 is System Manager and has access to all functions except setting user pass-codes

User 3-20 are operators, Operators have access to the light the flame, Read, AutoRead and AFHS functions to perform tests, view and print results and do calibration corrections.



Operation

From the factory the Admin pass-code is set to "1" and all others "0". On power up a flash screen indicates firmware and hardware version numbers.

BIO-AV
Flame Photometer

F3.20A32 H3.00 02.10

On pressing any key the user is prompted for a user ID and then pass-code.

User ID (1-20):

User ID (1-20): 1
Passcode (1-8 dig)
:

+ Features

-Prevent unauthorised access

-Aids in compliance to GLP and 21 CFR Part 11

-User IDs printed and logged against reports and calibrations

-LIMS integration

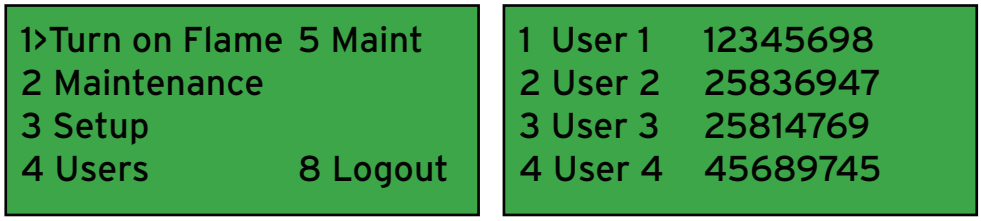
-Supports AFHS

-Enhance internal SOPs

-Audit trail monitoring

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After login as Admin item *4>Users* on the menu provides access to the User set-up screen; this screen shows a list of users with their pass-codes.



A user may be selected by using the arrow keys to move the cursor and then pressing Accept.
A new pass-code may then be entered for that user, a pass-code of "0" signifies user not allocated; a user can be effectively deleted by entering a "0" as the pass-code.
Selecting *8>Logout* will return to the login screen

Output to LIMS

Each result is output from the USB port as it is taken and the latest calibration details are output each time a calibration or correction is completed.
All Autoread samples will have a trayId of "0".

Result Output Format:*

```
<result>
	<header dateTime="DD/MM/YYYY HH:mm" trayId="2" sampledId="12" type="A" userId="12"/>
	<readings>
		<reading ion="Na" conc="140.5" units="mmol/l" raw"291458"/>
		<reading ion="K" conc="5.21" units="mmol/l" raw="84123"/>
	</readings>
</readings>
```

Calibration Output Format:*

```
<calibration>
	<instrument title="BWB Flame Photometer - BWB - BIO" serialNo ="20191234"
firmware="V3.20"/>
	<header dateTime="DD/MM/YYYY HH:mm"/>
	<ions>
		<ion name="Na" units="mmol/l" dilutionRatio="100" calibUser"2" correctUser="12"/>
			<points>
				<point conc="0.0" raw="20345"/>
				<point conc="140.0" raw="2911232"/>
			</points>
		</ion>
		<ion name="K" units="ppm" dilutionRatio="10" calibUser="1" correctUser="14"/>
			<points>
				<point conc="0.0" raw="3045"/>
				<point conc="5.10" raw="83245"/>
			</points>
		</ion>
	</ions>
</calibration>
```

*Note: Only carriage return are output at the end of each line - new lines and tabs have been included above to improve readability.

Flash - XP Comparison Information

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www.bwbtech.com

	XP	FL-N	FL-K	FL-L	FL-C	FL-2	FL-3	FL-4
Sodium analysis	✓	✓	✗	✗	✗	✓	✓	✓
Potassium analysis	✓	✗	✓	✗	✗	✓	✓	✓
Lithium analysis	✓	✗	✗	✓	✗	✗	✓	✓
Calcium analysis	✓	✗	✗	✗	✓	✗	✗	✓
Barium analysis	✓	✗	✗	✗	✗	✗	✗	✗
Simultaneous detection	Na, K, Li, Ca, Ba	Na	K	Li	Ca	Na, K	Na, K, Li	Na, K, Li, Ca
Calibration points per Ion (excl blank)	Up to 10	Up to 10	Up to 10	Up to 10	Up to 10	Up to 10	Up to 10	Up to 10
Calibration curves (Saved methods)	1	1	1	1	1	1	1	1
Multi element calibration curve	✓	✓	✓	✓	✓	✓	✓	✓
Internal results storage	✓	✓	✓	✓	✓	✓	✓	✓
AFHS Link	Option	✗	✗	✗	✗	✗	✗	✗
Integrated thermal printer	Option	✗	✗	✗	✗	✗	✗	✗
0 - 2.5v Chart output	✓	✗	✗	✗	✗	✗	✗	✗
4 - 20mA Output	Option	✗	✗	✗	✗	✗	✗	✗
Collection cup	Option	✗	✗	✗	✗	✗	✗	✗
Internal compressor	✓	✓	✓	✓	✓	✓	✓	✓
Ready to use out of the box	✓	✓	✓	✓	✓	✓	✓	✓
PC Software included	✓	✓	✓	✓	✓	✓	✓	✓
PDF Report generation	✓	✓	✓	✓	✓	✓	✓	✓
CSV Report generation	✓	✓	✓	✓	✓	✓	✓	✓
IQ OQ PQ possible	✓	✓	✓	✓	✓	✓	✓	✓
IQ OQ PQ kit included	✗	✗	✗	✗	✗	✗	✗	✗
Multiple gas selection	✓	✓	✓	✓	✓	✓	✓	✓

BWB Technologies

Li 3 Na 11 K 19 Ca 20 Ba 56 XP

Technical Data

Just Add Gas

Sample rate
2 - 3.5ml/min

Tubing materials
Silicone and Tygon®

Required desk space for instrument
50cm (H) x 45cm (W) x 45cm (D)
For Safety reasons the Flame photometer requires 1m of unobstructed space above to allow dissipation of heat from the chimney.

Instrument size
XP Flame Photometer
51cm (H) x 39cm (W) x 35cm (D)
(20in x 15in x 14in)
Shipping
65cm (H) x 49cm (W) x 43cm (D)
(25in x 19in x 17in)

Weight
XP Flame Photometer - 14kg (30.8lbs)
Shipping - 25kg (55.1 lbs)

Optimal range
Single point calibration
Na 0.1 - 60ppm
K 0.05 - 100ppm
Li 0.05 - 50ppm
Ca 1.0 - 100ppm
Ba 1.6 - 100ppm

Multi Point Calibration
Na 0.1 - 1000ppm
K 0.05 - 1000ppm
Li 0.05 - 1000ppm
Ca 1.0 - 1000ppm
Ba 5.0 - 3000ppm

Reproducibility
<1% Coefficient of variability for 20 consecutive samples over ten minutes at concentrations of 100ppm or less. (After instrument stabilisation).

Specificity
Na/K/Li = <1% to each other when equal in concentration at <100ppm

Limit of detection (LOD) and limit of quantification (LOQ)

LOD	LOQ
Na - 0.03ppm	Na - 0.1ppm
K - 0.02ppm	K - 0.05ppm
Li - 0.02ppm	Li - 0.05ppm
Ca - 0.3ppm	Ca - 1ppm
Ba - 1.6ppm	Ba - 5.0ppm

Time to stability
Less than 15 seconds after sample is introduced into the flame

Ions measured
Simultaneous measurement of Na, K, Li, Ca, Ba

Interfaces
USB
0-2.5 volt output (based on sample concentration linked to element of users choice)
Optional 4-20mA output in place of the above
Optional integrated printer
.csv and .pdf generated reports and files via FP-PC software

Recommended minimum warm up time based on ambient temperature
21ºC - 25 minutes

Power requirements
100V - 250V AC at 50 or 60Hz automatically selecting

Fuel requirement
Propane, Butane or Natural Gas*
regulated to 19Bar. Flow rate of 0.4l/min
*with modifications. BWB Technologies recommends either Propane or Butane for optimum results.

Readout
LCD, four line, alpha numeric, back lit.

+ WHAT'S IN THE BOX

- Power lead to suit region
- Gas hose
- Aspiration kit
- Manuals covering all aspects of the XP FP
- Quick start guides
- Warranty registration form
- Certificates of analysis for all provided fluids
- Material safety data sheets for all provided fluids
- The FP-PC software installation USB
- 150ml of BWB recommended cleaning solution
- 150ml calibration fluids for all 5 elements at 10,000ppm
- 100ml conical flask with screw lid
- 100 disposable sample cups
- 1L of de-Ionised water
- 2m of waste tube
- 10 x 10ml pipettes
- 10 x 1ml pipettes



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Inkarp Instruments Pvt Ltd (Head Office)

Plot No. 5A/10-11, IDA Nacharam Road No. 1, Nacharam - Chilka Nagar Road, Hyderabad - 500076

Email: sales@inkarp.co.in | service@inkarp.co.in | Ph. No.: +91 73311 80596 I Telephone: +91 40 2717 2293